

Multi



Enhancing network security and infrastructure at Multi AGT

By

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202003476

Project Thesis

Date:

Declaration

I confirm that this submission is entirely my own work. To the best of my knowledge, it does not include any content that has been previously published, authored by someone else, or submitted for the fulfillment of any degree or diploma at this or any other institution of higher education, unless explicitly referenced and properly acknowledged within the document.

Signature Ali Khalid

Name Ali Khalid

date:

Abstract

This project aims to bolster the security, stability, and operational performance of Multi Agencies General Trading's network infrastructure by introducing several critical enhancements. The core goals involve the deployment and configuration of a next-generation firewall to fortify network defenses, the integration of a secondary ISP router for uninterrupted internet connectivity, and the implementation of Azure Active Directory (Azure AD) for centralized cloud-based management of users and devices. The firewall will apply advanced protection protocols to guard against cyber threats while ensuring secure and efficient data flow. The backup ISP router will act as a failover option, minimizing service interruptions in the event of a primary connection failure. Furthermore, Azure AD will centralize identity and access management, simplifying security and authentication processes across the organization.

To visually demonstrate these upgrades, the project will include a Packet Tracer simulation that mirrors the company's actual network layout. This simulated environment will feature EtherChannel for improved link aggregation, OSPF for dynamic routing capabilities, VLANs with inter-VLAN routing for structured network segmentation, and DHCP for automatic IP assignment. NAT will be configured to enable secure internet access, and wireless access points will be included to extend internal connectivity.

Looking ahead, future upgrades may include refining the firewall rule set to enhance threat monitoring and mitigation, increasing network resilience through additional redundancy mechanisms, and strengthening Azure AD by introducing more granular access controls, role-based authorizations, and multi-factor authentication. Automating user onboarding processes and enforcing consistent security settings across all endpoints will also contribute to greater efficiency and protection. Collectively, these improvements will help establish a robust, scalable, and well-managed IT infrastructure capable of supporting both current operations and future expansion.

Acknowledgments

I would like to extend my heartfelt appreciation to Mr. Shahbaz Ali for his invaluable support and guidance throughout the duration of this project. His expertise and clear perspective at each stage greatly contributed to overcoming challenges and deepening my technical understanding. His structured mentorship, combined with the encouragement to explore solutions independently, significantly enhanced my learning journey. I am truly thankful for the time and dedication he invested in my academic and professional growth.

My sincere thanks also go to my academic advisor, Ms. Ruqaya Yusuf, whose consistent support and insightful feedback played a crucial role in the success of this project. Her thoughtful guidance during our consultations boosted my confidence in handling complex tasks and helped sharpen both my technical skills and overall professional development.

I am equally grateful to Mr. Hussain Ferdousi, my company mentor at Multi Agencies General Trading, for sharing his industry knowledge and practical experience. His guidance was key to understanding the real-world applications of networking concepts and implementing solutions effectively in a professional setting.

I deeply appreciate the mentorship and encouragement I received from all of you. It has been a privilege to work under your guidance, and I am inspired by your commitment to supporting my growth.

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1. Introduction

1.1. Project Rationale

In a business environment where network reliability and security are vital—particularly for a general trading company combined with a supermarket—this project aims to optimize the IT infrastructure of Multi Agencies General Trading (MAGT). As the company continues to grow and attract new customers, the existing network must evolve to accommodate increased demand, ensure high availability, and bolster cybersecurity measures.

Challenges such as inconsistent connectivity, security vulnerabilities, and limited centralized management pose serious risks to operational efficiency and business continuity. A robust and well-structured network is essential to maintain seamless day-to-day operations, protect sensitive data, and enable secure communication across departments. Without sufficient security and redundancy, businesses face increased exposure to data breaches, system failures, and operational disruptions—events that could lead to financial losses and a decline in productivity.

1.2. Project Objectives

This project is designed to address the aforementioned challenges by implementing a resilient network architecture. The proposed upgrades include:

- Deployment of a next-generation firewall to enhance security.
- Configuration of a backup ISP router to provide internet redundancy.
- Integration of a cloud-based Azure Active Directory for centralized user and device management.

These improvements aim to streamline IT operations, reduce downtime, and strengthen the overall security posture. By modernizing its network infrastructure, MAGT will be better equipped to support future expansion, safeguard critical business data, and maintain a secure and reliable IT environment.

1.3. Prior Work

The foundation of this project builds upon existing research and industry practices related to network design, ISP redundancy, and cybersecurity standards. Past research into ISP connectivity has explored various models for multi-ISP environments. In the case of MAGT, a dual-ISP setup is sufficient, focusing on redundancy and routing protocol efficiency—particularly relevant in the context of Bahrain and the wider Middle East region.

Research on network security emphasizes the importance of firewalls, VPNs, Access Control Lists (ACLs), and Network Address Translation (NAT) in securing data across private and public networks. However, MAGT's existing infrastructure has yet to fully adopt or configure all of these protocols.

While the literature provides valuable theoretical insights, it often lacks practical guidance on integrating multiple technologies within a small-to-medium-sized business network. Many studies either focus on single-ISP configurations or large enterprise-level environments, rarely addressing the challenges of managing connections across multiple ISPs at the SME level. Specific issues include maintaining secure, efficient internal communication while simultaneously enforcing external security controls.

Furthermore, there is limited documentation on the real-world implementation of scalable security solutions, such as configuring firewalls in a multi-ISP setup or deploying cloud-based Active Directory for centralized management. This project seeks to bridge that gap by providing detailed documentation of the integration process, including security configurations and protocol optimizations. The result will serve as a practical guide for businesses aiming to achieve secure, resilient, and scalable connectivity within a dual-ISP environment.

1.4. Proposed solution

To enhance the connectivity, security, and performance of the company's network, this project will deliver a structured and resilient network solution tailored to Multi Agencies General Trading. The approach includes:

- Implementing network redundancy through a backup ISP router, ensuring uninterrupted connectivity and minimizing data loss during outages.
- Securing internal and external access to the network by applying strict access control policies and deploying advanced security measures.
- Improving internal network efficiency through structured traffic management, with the integration of DHCP services into the firewall system. This change replaces the previously ISP-managed DHCP service, which did not meet MAGT's operational standards.

The firewall will enforce organization-wide security policies, protecting the network from external threats while maintaining operational integrity. These measures will support a scalable IT environment capable of adapting to business growth while maintaining high availability and system resilience.

1.5. Objectives/Tasks:

- Ensure seamless internal communication across the network where needed.
- Maintain stable external connectivity by enabling automatic failover between service providers in the event of an ISP outage.
- Deploy a centralized management system to streamline IT administration tasks and device control.
- Implement essential security features on the firewall tailored to the specific needs and risks of the company.
- Enforce application control policies to block unauthorized software and services that could pose threats or reduce productivity.
- Continuously monitor and log network activity to detect anomalies and proactively respond to potential security threats.

1.6. Description of the report

This thesis provides a comprehensive analysis of the initiative to enhance the network infrastructure of Multi Agencies General Trading. The report begins with the Background Work section, which introduces the foundational technical concepts and outlines the technologies selected to support the project's objectives.

The Requirements section defines the methods used to assess the existing environment and identify key needs, including security, scalability, and operational efficiency. It highlights both the current limitations and the goals for improvement.

The Design and Methodology section outlines the architectural decisions and strategic approaches taken to ensure the new infrastructure aligns with business goals, emphasizing reliability, flexibility, and future growth potential.

In the Implementation section, the report details how the proposed solution was put into action. This includes system configurations, integration of new devices and services, and the application of enhanced security protocols.

The Testing section evaluates the effectiveness of the deployed solution through a series of performance checks, including hardware load testing, threat simulations, and connectivity assessments. These tests validate that the system meets its intended goals.

Finally, the Discussion and Conclusion section summarizes key achievements, challenges encountered, and lessons learned throughout the project. It also provides recommendations for future improvements in network security and operational efficiency, along with general IT usage guidelines for company staff to support long-term sustainability.

2. Background

2.1. Related theory

For modern businesses—particularly those reliant on a single Internet Service Provider (ISP)—effective network security and infrastructure design are critical components of operational resilience. Contemporary research highlights the importance of integrating firewalls, backup ISP connectivity (commonly enabled through technologies like SD-WAN), and centralized user authentication mechanisms to enhance both reliability and protection. Solutions such as FortiGate firewalls are frequently cited for their ability to control traffic flow, enforce access policies, and guard against cyber threats. Notably, organizations that implement next-generation firewalls (NGFWs) often benefit from improved visibility into network activity and stronger defenses against evolving security risks.

Network redundancy is another cornerstone of robust infrastructure. Studies examining dual-ISP environments underscore how backup connections significantly reduce service disruptions and uphold business continuity. While dynamic routing protocols like Border Gateway Protocol (BGP) allow for automatic and seamless failover between ISPs, many small and medium enterprises opt for more manageable solutions like static failover, which still provide effective redundancy with simplified configuration. Both approaches were reviewed and considered in the context of the Multi Agencies General Trading project.

On the management front, the shift towards cloud-based solutions has brought identity services such as Azure Active Directory (AAD) to the forefront. AAD offers streamlined access control, identity governance, and device management, particularly suited for businesses embracing modern cloud ecosystems. Unlike traditional on-premises Active Directory deployments, Azure AD facilitates more flexible and scalable identity management. Research supports that integrating AAD with local infrastructure enhances security and administrative oversight while aligning with current cloud-first strategies. This project leverages these concepts to deliver a network environment tailored to Multi Agencies General Trading, built with an emphasis on stability, centralized control, and secure scalability.

2.2. Project Technologies

Microsoft Azure	Microsoft Azure is a comprehensive cloud services platform offering capabilities in computing, networking, storage, and identity management. In this project, Azure was utilized to host a cloud-based Active Directory, enabling centralized
------------------------	---

 FIGURE 1 - MICROSOFT AZURE	administration of user identities, permissions, and access control across the organization's digital infrastructure.
Backup ISP router  FIGURE 2 - BACKUP ISP ROUTER	A backup ISP router serves as a contingency device configured to automatically reroute internet traffic in the event of a failure in the primary connection. For the Multi AGT setup, this was implemented to safeguard continuous connectivity and minimize potential disruptions to mission-critical business services.
FortiGate Firewall  FIGURE 3 - FORTIGATE FIREWALL	FortiGate, a next-generation firewall solution by Fortinet, delivers a range of advanced security functionalities including intrusion detection and prevention, secure VPN access, content filtering, and anti-malware defense. The project team selected FortiGate for its proven performance, intuitive management interface, and capacity to handle enterprise-level traffic with high reliability.
Azure Active Directory  FIGURE 4 - AZURE ACTIVE DIRECTORY	Azure Active Directory is Microsoft's cloud-based identity and access management platform. In this project, AAD was adopted to oversee user credentials, define access roles, and enforce security policies across organizational resources, thereby supporting a secure and unified identity framework for employees and administrators alike.

3. Methodology

3.1. Solution Design

As this project was implemented in a real business environment, strict company policies and a signed Non-Disclosure Agreement (NDA) prevented the disclosure of Multi Agencies General Trading's actual network infrastructure. To accommodate this, a simulated network was developed using Cisco Packet Tracer to provide a representative model of the proposed solution and demonstrate key design principles.

The simulated design features a two-floor structure, with each floor treated as a separate network segment. On the second floor, a NAT (Network Address Translation) router is connected to a Layer 3 switch. This switch is linked to an access layer switch configured with VLANs to logically separate the IT department and customer-facing services. Additionally, a wireless access point was included to support mobile and wireless device connectivity in this area.

The first floor of the topology includes another router connected to a Layer 3 switch, which distributes connectivity to three dedicated switches—each serving a different department: Finance, Human Resources (HR), and Sales. These departmental switches are also VLAN-configured to ensure proper segmentation and control of network traffic.

This Packet Tracer simulation was designed to fulfill the project's key requirements: internal segmentation, scalability, redundancy, and basic wireless coverage. It serves as a visual and functional reference for the kind of solution deployed in the real-world environment, without compromising the confidentiality agreement.

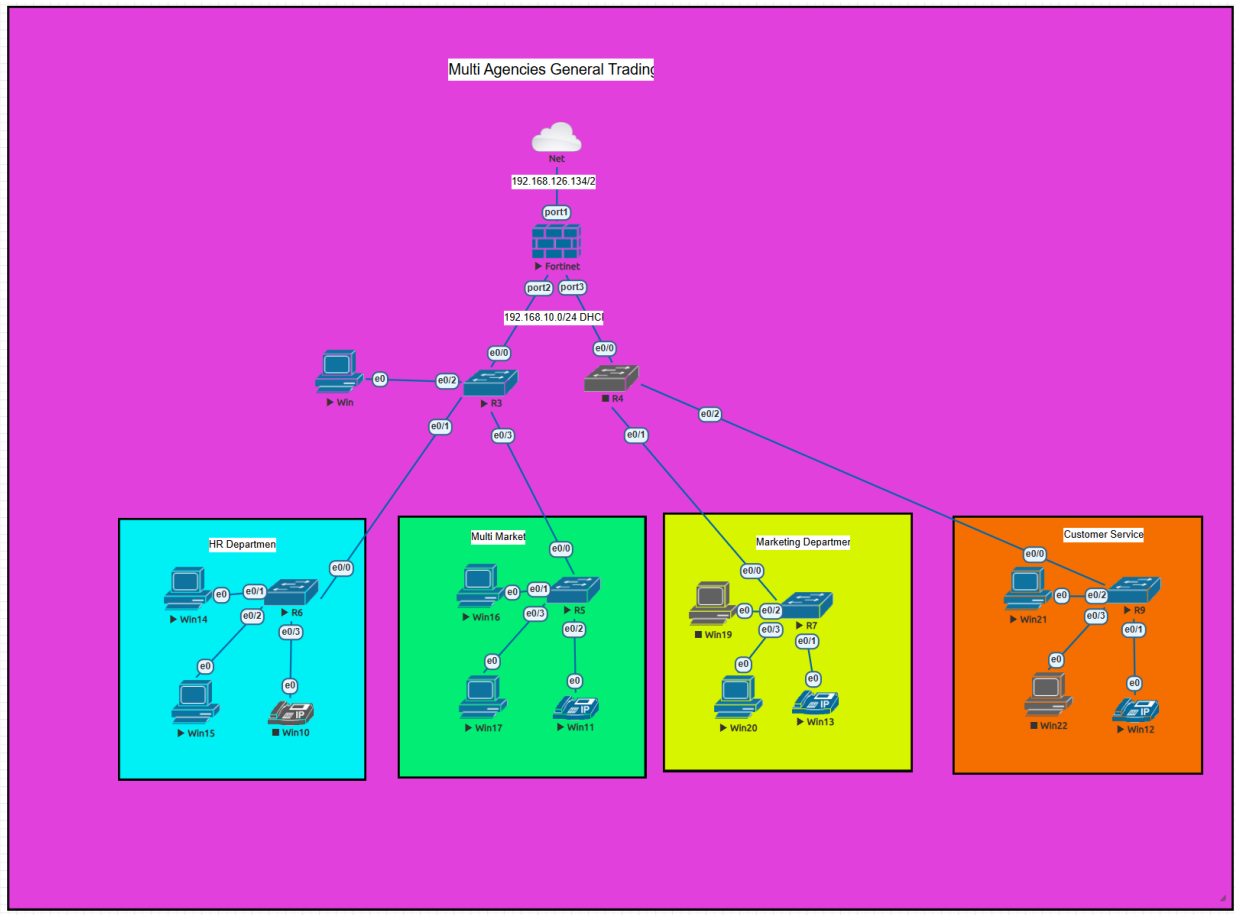


FIGURE 5 - SOLUTION DESIGN

3.2. Use Case Diagram

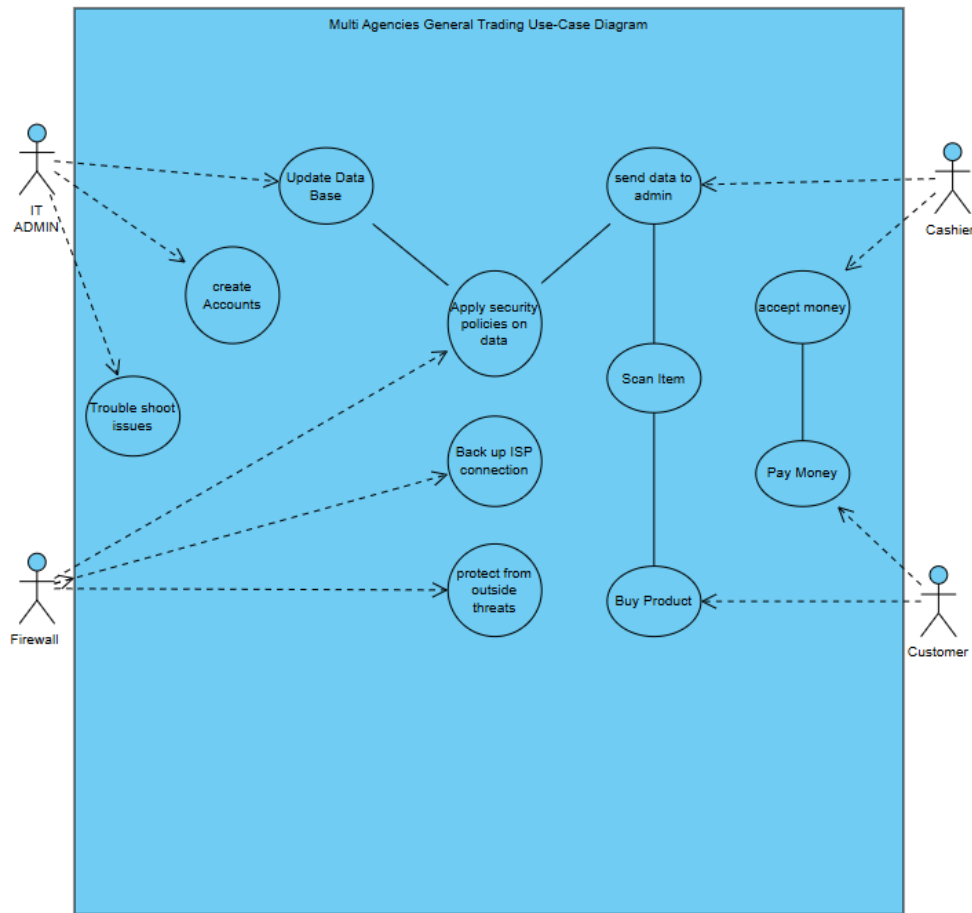
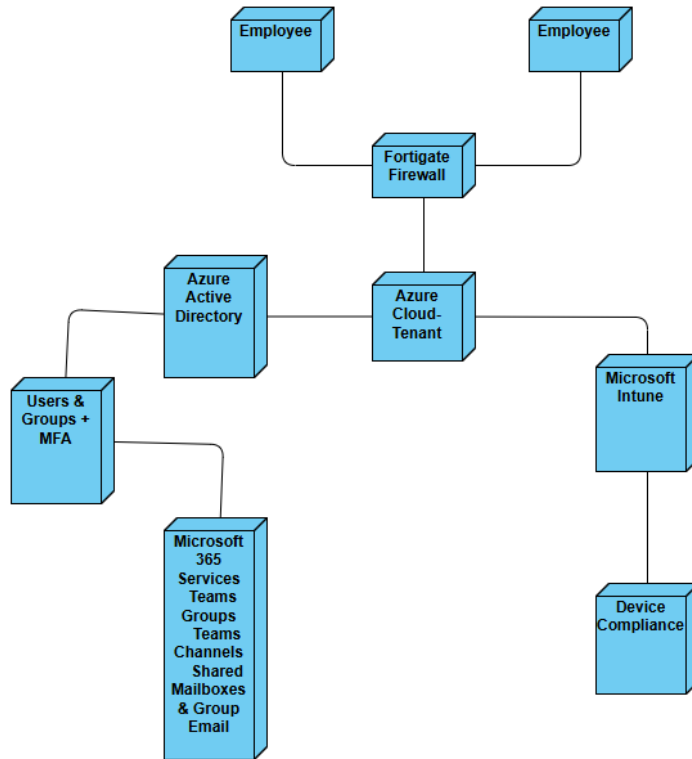


FIGURE 6 - USE CASE DIAGRAM

The Use case Diagram above different actors who all have different jobs at Multi AGT, IT Admin, Cashier, Customer, and firewall, they all work together within the company, the most technical role is the IT admin since he manages the firewall, as well as inserts the updated data into the database.

3.3. Deployment Diagram



3.4. Functional Requirements

Requirement	Priority
Deploy a firewall to secure internal and external traffic	High
Install and configure a secondary ISP router for redundancy	High
Integrate Azure Active Directory for user identity management	High
Enable DHCP for automated IP allocation	Medium

Apply basic IP-based segmentation for departmental isolation	Medium
Configure NAT for external internet access	Medium
Track and analyze traffic for potential security incidents	Low
Maintain logging and audit records for network activities	Medium
Implement email filtering to reduce malicious content	Low
Maintain documentation of the network topology and settings	Medium

TABLE 1 - FUNCTIONAL REQUIREMENTS

3.5. Non-Functional Requirements

Requirement	Priority
Maintain a network uptime rate of no less than 95%	High
Ensure the firewall handles at least 250 concurrent sessions	High
Automatic failover to backup ISP must occur within 30 seconds	High
Align system implementation with Bahrain's cybersecurity laws	Medium
DHCP lease configuration must optimize efficiency and reuse	Low
Email security should successfully detect 80% of spam/phishing	Medium
Network should scale to support 30% user growth over 3 years	Low
Ensure technical documentation remains clear and up-to-date	Low

TABLE 2 - NON-FUNCTIONAL REQUIREMENTS

3.6. Market Analysis

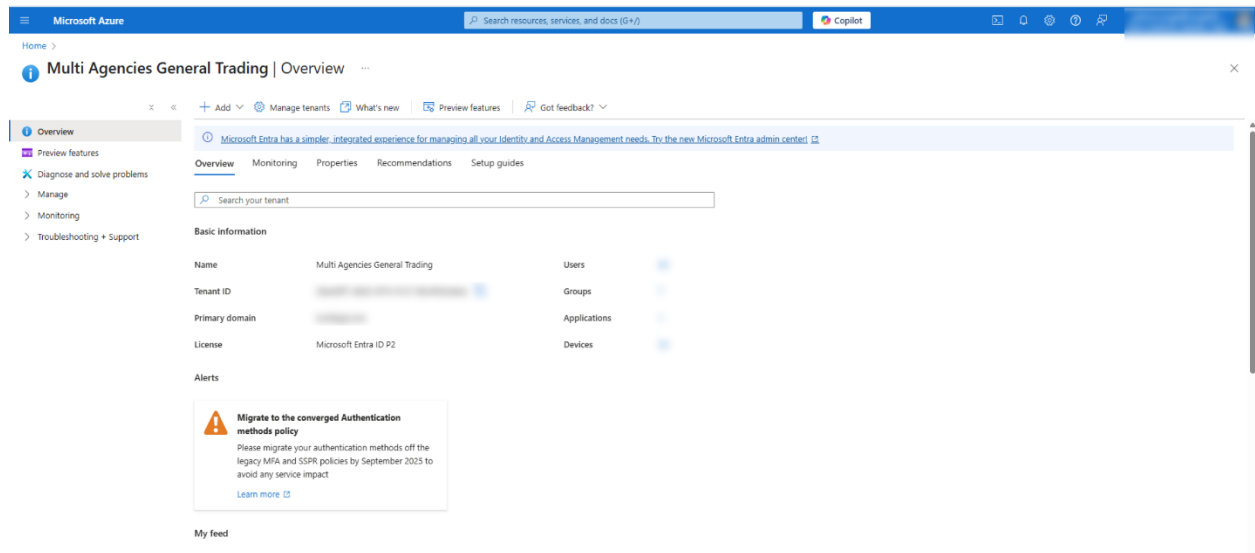
Feature	Fortinet	SonicWall	Cisco
Overview	High-performance firewall offering secure SD-WAN and	Enterprise-grade firewall suitable for SMBs, offering	Industry-standard solution with strong network integration

	deep threat protection.	advanced VPN and threat detection tools.	and enterprise-level security controls.
Servers & Workstations	Intel Xeon E5+, 16GB+ RAM, 1TB SSD	Customizable CPU, 8GB+ RAM, 500GB+ storage	Intel Xeon CPU, 16GB+ RAM, 1TB SSD
Software Tools	FortiManager, EVE-NG, integrated IDS/IPS	SonicWall NSM, threat intelligence platform, cloud console	Cisco Firewall Management Center, advanced IDS/IPS tools
Applications	FortiAnalyzer (analytics), FortiClient (endpoint security)	Capture ATP (cloud sandboxing), Global VPN Client	Cisco AnyConnect (VPN), Cisco Umbrella (DNS filtering)
Data Processing	Real-time inspection with automatic security updates	Mix of real-time and batch for deep threat analytics	Real-time processing with cloud-based threat response
Output Devices	Firewalls, Web Gateways, VPN appliances	Firewalls, Web filtering systems, VPN concentrators	Firewalls, Secure Email Gateways, IPS systems
Input Devices	GUI/CLI consoles, monitoring dashboards	Web UI, Command-line access, remote security tools	Smart Dashboard, web-based interfaces, CLI
Storage Options	Local storage (1TB+), threat logs, compliance-ready databases	Cloud or on-premises log storage depending on sensitivity	Hybrid storage with threat intelligence maintained both locally and in the cloud

TABLE 3 - MARKET ANALYSIS

4. Implementation

Note: sensitive company data has been blurred as requested by my mentor in the IT department.



After logging in Microsoft active directory a tenant was created with the name multi agencies general trading with its primary domain being multiagt.com. This screenshot shows the Azure Active Directory tenant created for the project. A tenant is a dedicated environment within Azure that provides identity and access management. It serves as the foundation for managing users, groups, and devices within the organization. Also in the screen shot the domain name is visible, what a domain does is allow easier management for the admin of user accounts which were created with this domain.



[Home](#) > [Multi Agencies General Trading | Users](#) > [Users](#) >

Create new user

Create a new internal user in your organization

[Basics](#) [Properties](#) [Assignments](#) [Review + create](#)

Create a new user in your organization. This user will have a user name like alice@contoso.com. [Learn more](#)

Identity

User principal name * @ [Domain not listed? Learn more](#)

Mail nickname *

☒ Derive from user principal name

Display name *

Password * [Auto-generate password](#)

Account enabled ☒

This screenshot shows the create new user page for the tenant, where name and display name are written based on employee identity card, a password is auto generated and the option to enable the account for the employee as well.

[Home](#) > [Multi Agencies General Trading | Users](#) > [Users](#) >

Create new user

Create a new internal user in your organization

[Basics](#) [Properties](#) [Assignments](#) [Review + create](#)

Identity

First name

Last name

User type

Authorization info [+ Edit Certificate user IDs](#)

Job Information

Job title

Company name

Department

Employee ID

Employee type

Employee hire date

Office location

Manager [+ Add manager](#)

Contact Information

Street address

City

[Review + create](#)

[< Previous](#)

[Next: Assignments >](#)

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home > Users

Users | Audit logs

Multi Agencies General Trading

Download Refresh Columns Got feedback?

This view will soon be replaced with a view that includes custom security attribute logs, infinite scrolling, and column reordering. Try out our new audits preview. →

Date: Last 1 month Show dates as: Local Service: All Category: UserManagement Activity: All Add filters

Date	Service	Category	Activity	Status	Status reason	Target(s)	Initiated by (actor)
06/05/2025, 10:16:34	Core Directory	UserManagement	Update user	Success			
03/05/2025, 16:25:56	Core Directory	UserManagement	Update user	Success			
30/04/2025, 13:08:29	Core Directory	UserManagement	Update user	Success			
29/04/2025, 18:10:42	Core Directory	UserManagement	Update user	Success			
29/04/2025, 18:10:42	Authentication Methods	UserManagement	User registered all required sec...	Success	User registered all required sec...		
29/04/2025, 18:10:35	Authentication Methods	UserManagement	User registered security info	Success	User registered Authenticator ...		
29/04/2025, 18:10:34	Core Directory	UserManagement	Update user	Success			
29/04/2025, 18:10:21	Authentication Methods	UserManagement	User started security info regist...	Success	User started the registration for...		
29/04/2025, 18:10:10	Core Directory	UserManagement	Update user	Success			
29/04/2025, 18:10:10	Core Directory	UserManagement	Update user	Success			
29/04/2025, 17:42:04	Core Directory	UserManagement	Update user	Success			
29/04/2025, 16:41:08	Self-service Password Manage...	UserManagement	Reset password (self-service)	Success			
29/04/2025, 16:41:08	Core Directory	UserManagement	Update StorefreshTokenValidit...	Success			

this screen shot shows all the audit logs from the past month with dates service type, activity and its categories, and finally their status. This helps the IT department at multi agencies general trading, to have an overview of what the users are doing and where they need help in case of a technical problem.

Microsoft Azure

Home > Multi Agencies General Trading | Devices > Devices | Overview

All devices

Download devices Refresh Manage view Enable Disable Delete Manage Preview features Got feedback?

Search by name or device ID or object ID Add filters

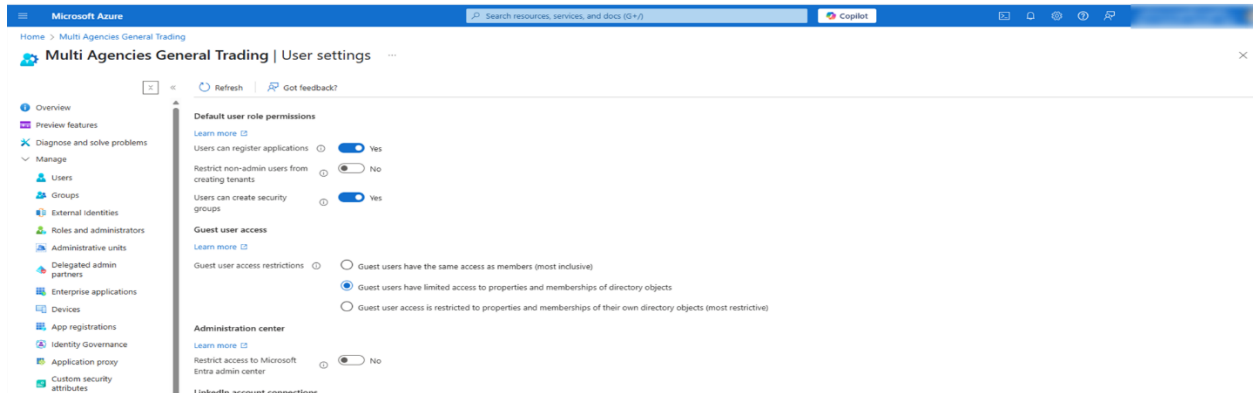
34 devices found

☐	Name	Enabled	OS	Version	Join type	Owner	MDM	Security settings m...	Compliant	Registered	Activity
☐		Yes	Windows	10.0.26100.3775	Microsoft Entra reg...		Microsoft Intune	Microsoft Intune	Yes	28/10/2024, 12:00	04/05/2025, 07:...
☐		Yes	Windows	10.0.26100.3476	Microsoft Entra reg...		Microsoft Intune	Microsoft Intune	No	14/11/2024, 11:20	25/05/2025, 06:...
☐		Yes	Windows	10.0.26100.3775	Microsoft Entra reg...		Microsoft Intune	Microsoft Intune	Yes	06/11/2024, 12:29	02/05/2025, 22:...
☐		Yes	Windows	10.0.26100.3775	Microsoft Entra reg...		Microsoft Intune	Microsoft Intune	Yes	14/04/2025, 14:00	02/05/2025, 16:...
☐		Yes	Windows	10.0.22631.4460	Microsoft Entra reg...		None	N/A	No	27/10/2024, 09:50	27/04/2025, 09:...
☐		Yes	Windows	10.0.26100.3775	Microsoft Entra reg...		Microsoft Intune	Microsoft Intune	Yes	13/11/2024, 08:20	27/04/2025, 08:...
☐		Yes	Windows	10.0.26100.3775	Microsoft Entra reg...		None	N/A	N/A	29/04/2025, 16:55	29/04/2025, 16:...
☐		Yes	Windows	10.0.26100.3775	Microsoft Entra joi...		Microsoft Intune	Microsoft Intune	Yes	09/04/2025, 14:35	28/04/2025, 01:...
☐		Yes	Windows	10.0.26100.3194	Microsoft Entra reg...		None	N/A	N/A	04/03/2025, 12:50	04/05/2025, 08:...
☐		Yes	Windows	10.0.26100.1742	Microsoft Entra joi...		None	N/A	N/A	23/04/2025, 08:42	23/04/2025, 08:...
☐		Yes	iOS	17.7.2	Microsoft Entra reg...		None	N/A	N/A	12/01/2025, 11:48	29/04/2025, 12:...
☐		Yes	Windows	10.0.26100.3775	Microsoft Entra joi...		None	N/A	N/A	23/04/2025, 10:38	23/04/2025, 10:...
☐		Yes	Windows	10.0.26100.3775	Microsoft Entra joi...		Microsoft Intune	Microsoft Intune	Yes	09/04/2025, 10:58	09/04/2025, 10:...
☐		Yes	Windows	10.0.26100.3775	Microsoft Entra reg...		Microsoft Intune	Microsoft Intune	Yes	22/12/2024, 11:25	01/05/2025, 13:...

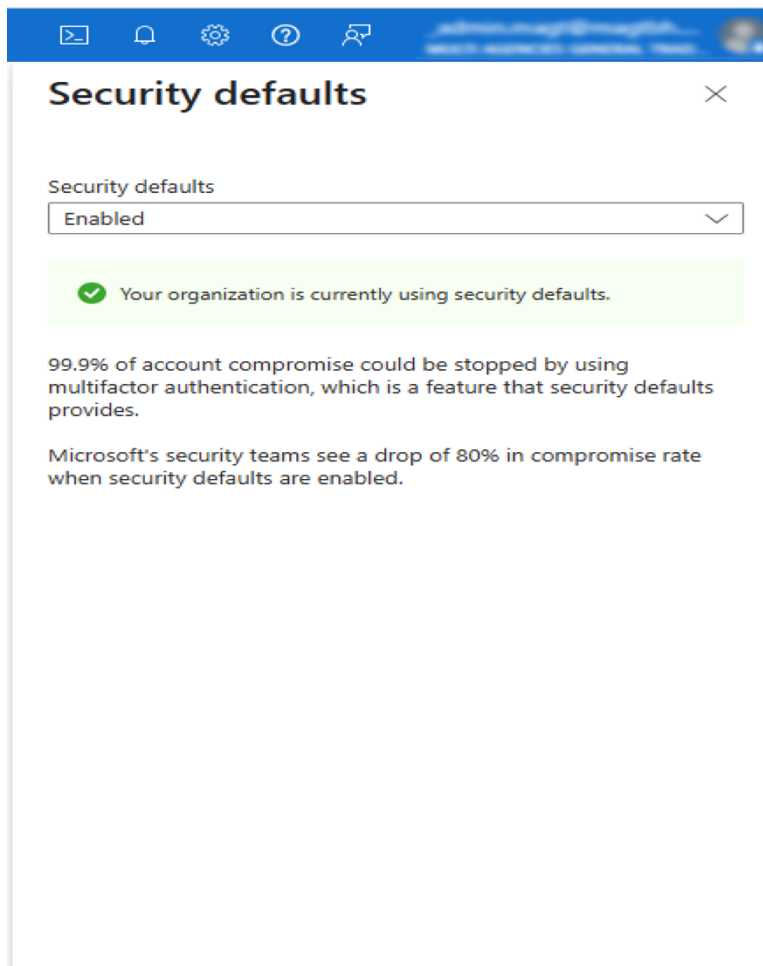
Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.
Go to Settings to enable Windows
Markup and share

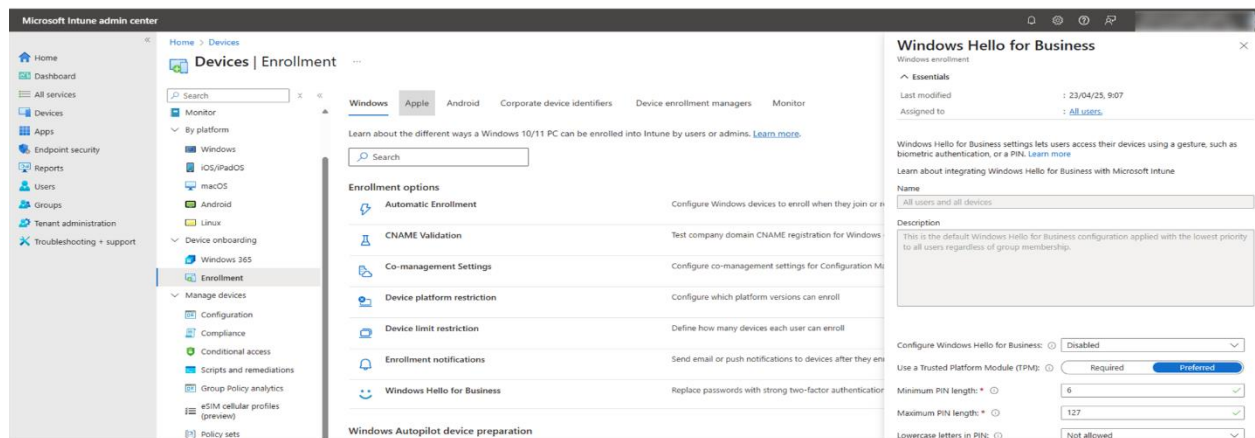
This screen shows all the devices that have logged in into the domain with their accounts that were created. After logging in into the domain the device registered inside azure as a company device so the admins have more management over the devices.



This screenshot shows the user settings which were changed after deployment of active directory which only shows privileges that account types have, such as members and admins.



The IT department also asked all accounts created for employees to be forced to use two factor authentication, which was enabled by enabling security defaults inside Microsoft Entra (azure active directory)



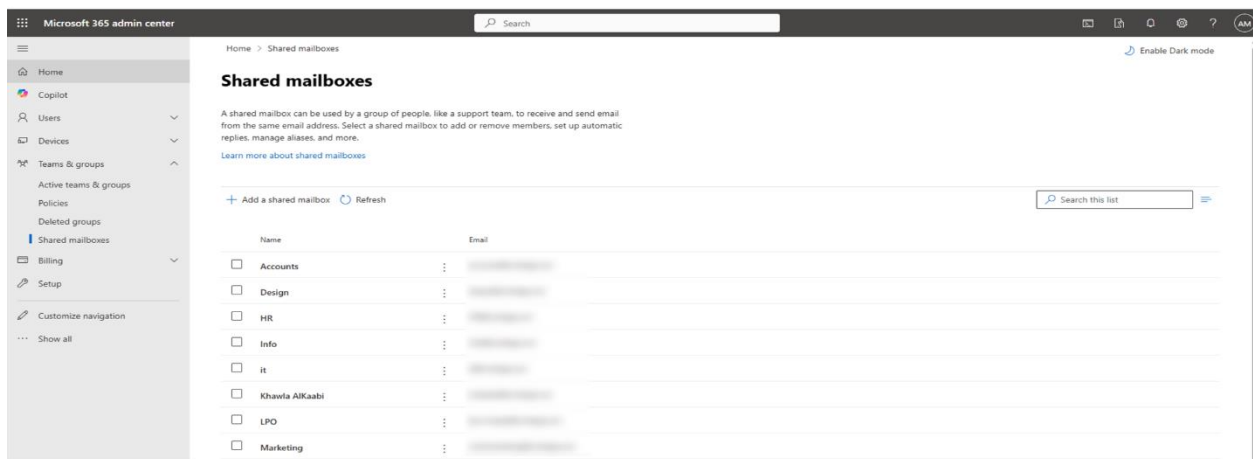
This screenshot shows Microsoft Intune, which is a Microsoft device and app management center, the IT department requested that new devices logging into the domain workplace, do not configure windows hello, which was added with windows 11, the screenshot shows that configure windows hello for business is disabled.

Device limit restrictions

Define how many devices each user can enroll.

Name	Device limit	Assigned
All users and all devices	5	Yes

This screenshot shows that all users inside the active directory have a limit of 5 user accounts which was also requested by the IT department.



The IT department asked for a shared mailbox, which basically means that a whole department like HR can use one email address to send and receive emails this screenshot shows that it was configured.

Microsoft Teams admin center

Search

Dashboard

Teams Premium

Teams

Manage teams

Teams settings

Teams policies

Team templates

Templates policies

Teams update policies

Teams upgrade settings

Users

Teams devices

Teams apps

Voice

Analytics & reports

Manage teams

Teams and channels are collections of people, content, and tools used for projects or outcomes within your organization. You can manage all teams and channels, and create new ones. Go to the [Admin center > Groups](#) to manage Microsoft 365 groups. [Learn more](#)

Users summary

40

Total users

40

Internal users

0

Guests

Quick guide to the new chats, teams and channels experience
[View](#)

Get essential resources to help your organization get started with the new experience.

+ Add
Edit
Archive
Delete
Renew
3 teams

Search for

✓	Name	Standard channels	Private channels	Shared channels	Privacy	Status	Description
	MA Multi Agencies General Trading	1	0	0	Public	Active	Multi Agencies General ...
	IT IT Team	1	9	1	Private	Active	Group for the IT depart...
	M Marketing	1	0	0	Private	Active	Marketing Team for mar...

Multi Agencies General Trading

Multi Agencies General Trading

MA

Team members

AF I M

Open in Teams

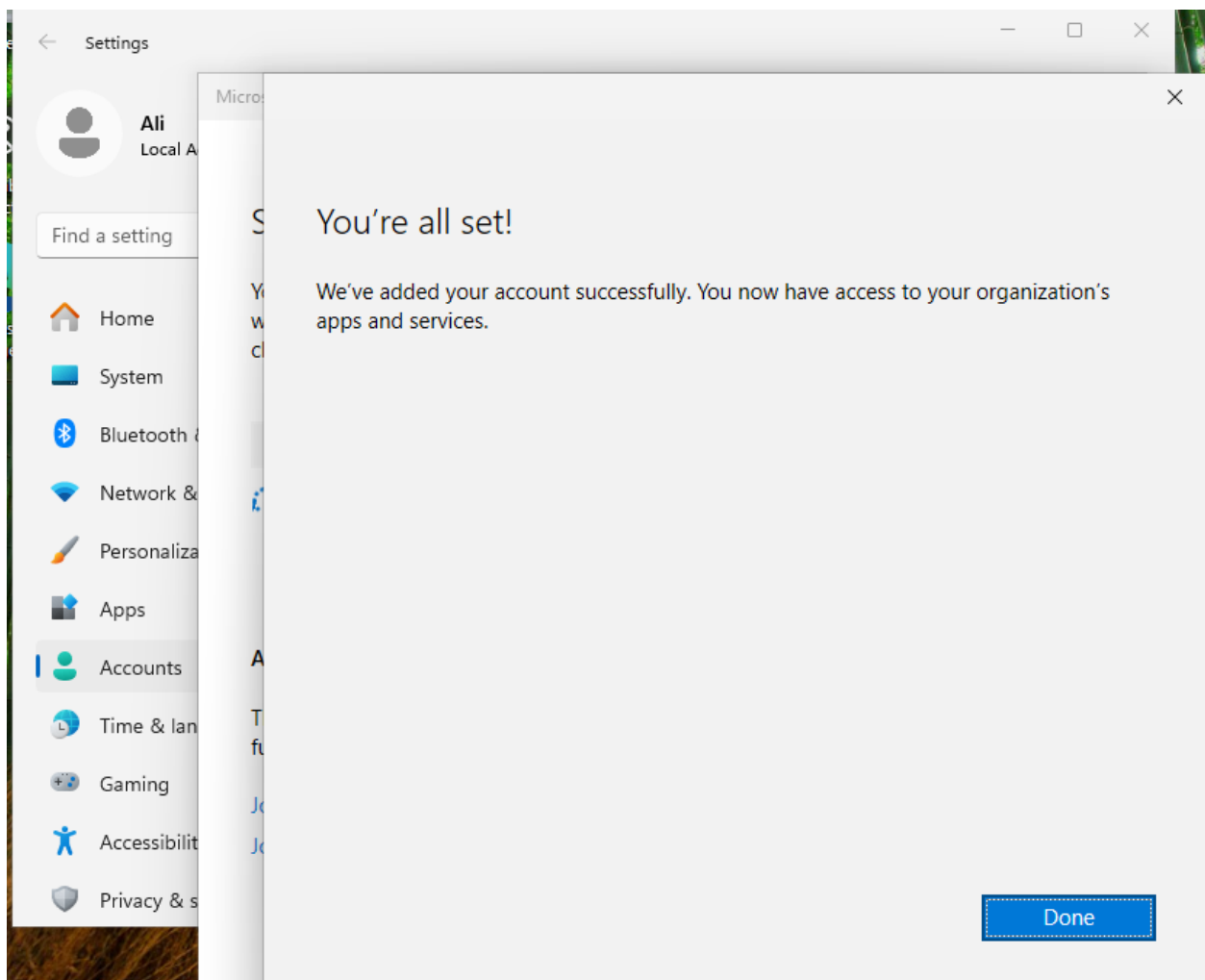
Send email

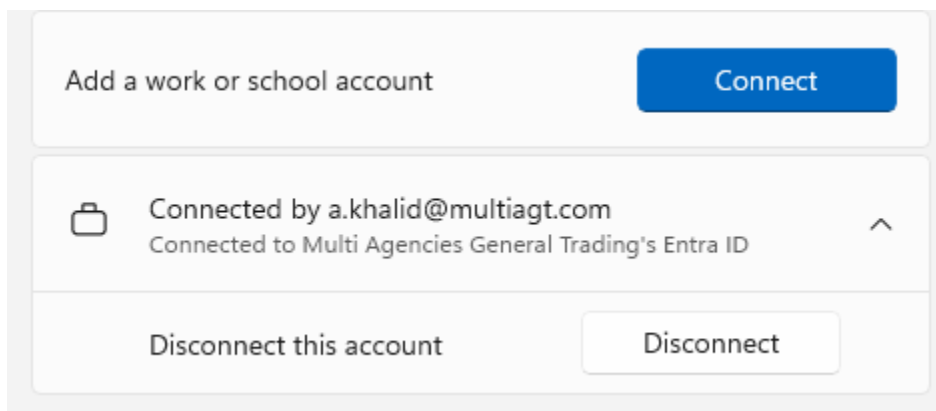
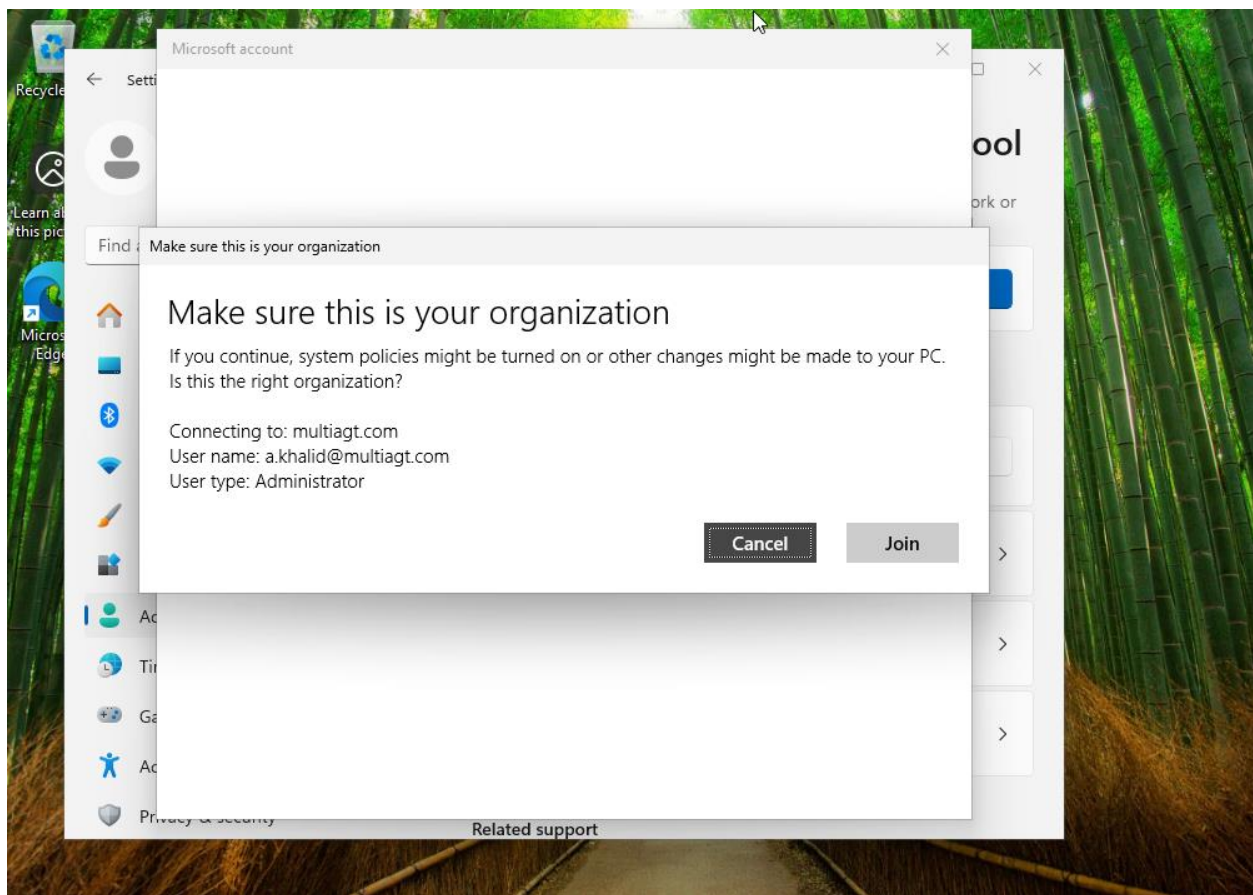
Privacy

Public

Email

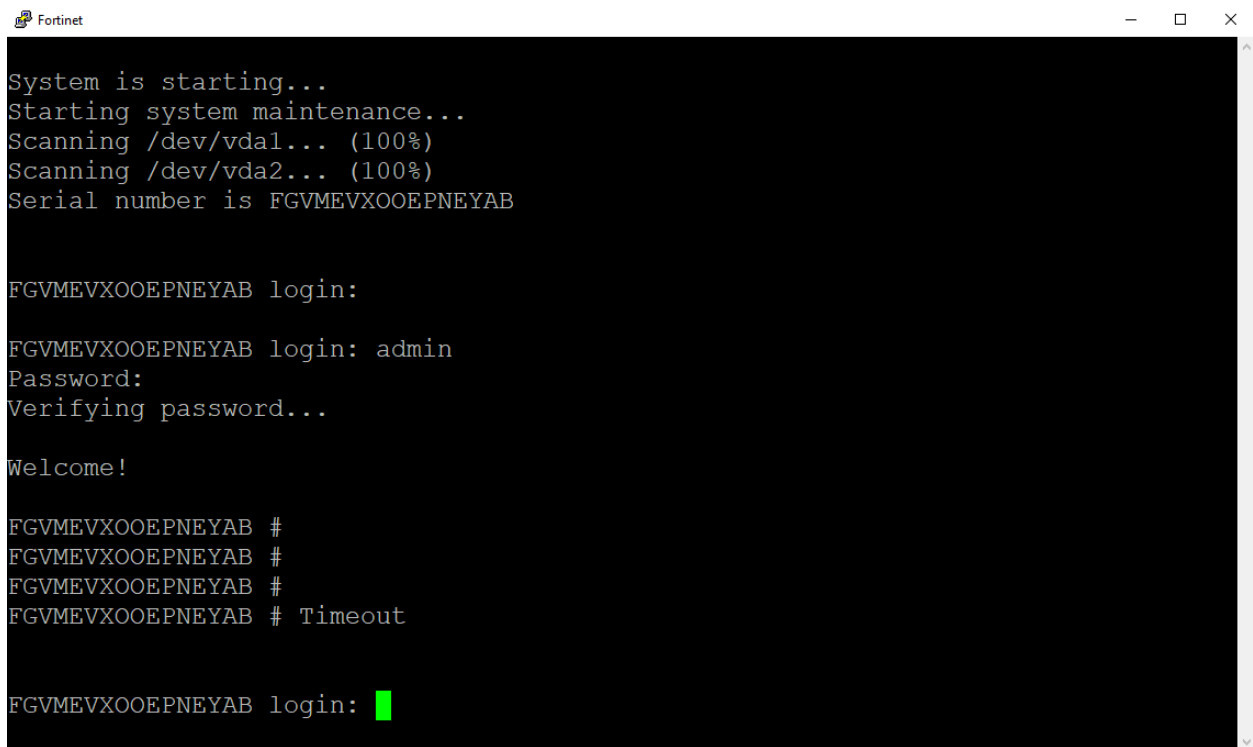
The screenshot above shows how Microsoft Teams Groups and Channels were used to improve communication and collaboration within the company. Teams Groups provided each department with a shared workspace that included chat, calendar, files, and email, making it easier to manage tasks and stay connected. Channels within each group were created to organize discussions by topic or project, allowing team members to communicate clearly and access relevant information quickly. This structure helped automate workflows, reduce confusion, and made it easier to track progress and updates.





The screenshots above show logging in into multiagt.com from a brand-new Win 11 device. After signing in the account all the policies and regulations provided by multi agt IT administrators will be run on this device which will allow easier management by the administrators.

FortiGate Configuration



```
Fortinet
System is starting...
Starting system maintenance...
Scanning /dev/vda1... (100%)
Scanning /dev/vda2... (100%)
Serial number is FGVMEVXOOEPNEYAB

FGVMEVXOOEPNEYAB login:

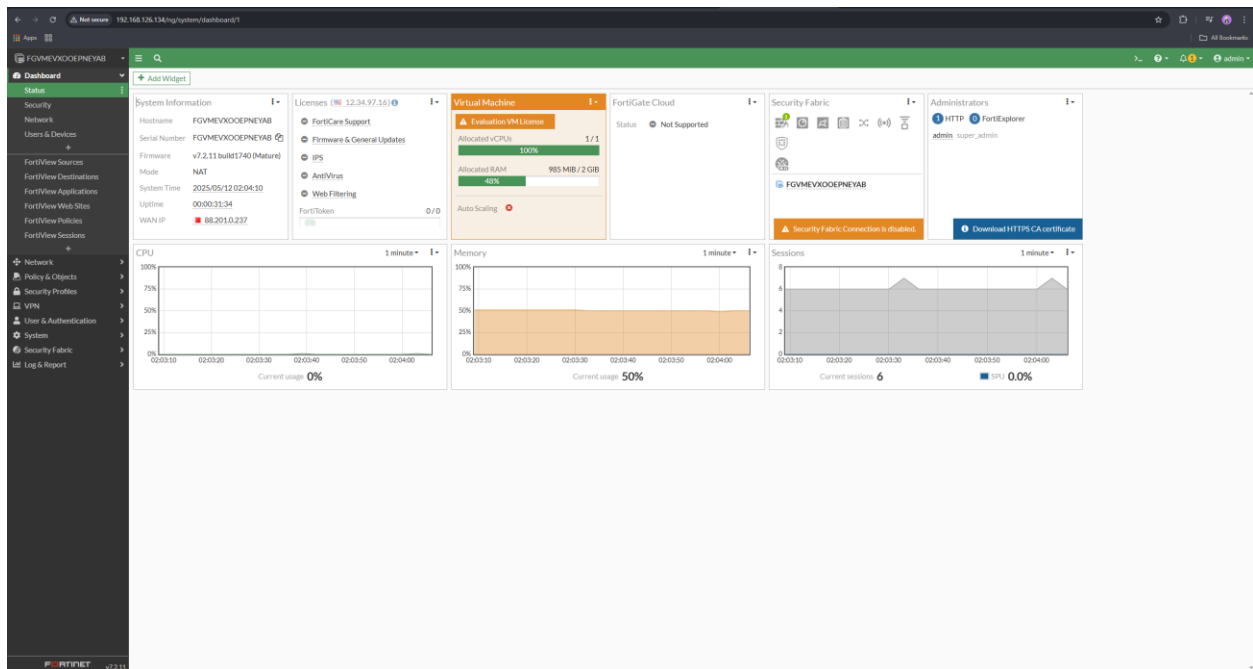
FGVMEVXOOEPNEYAB login: admin
Password:
Verifying password...

Welcome!

FGVMEVXOOEPNEYAB #
FGVMEVXOOEPNEYAB #
FGVMEVXOOEPNEYAB #
FGVMEVXOOEPNEYAB # Timeout

FGVMEVXOOEPNEYAB login: █
```

After installing the FortiGate image and starting up the firewall, i logged in with an administrator account with no password and updated the password. Running the show interface commands gives an Ip address for the FortiGate web interface



The image above shows the dashboard, and the status of the firewall

Edit Interface

Name

Wan (port1)

Alias

Wan

Type

Physical Interface

VRF ID

0

Role

WAN

Estimated bandwidth

0

kbps Upstream

0

kbps Downstream

Address

Addressing mode

Manual DHCP

Status

Connected

Obtained IP/Netmask

192.168.126.134/255.255.255.0

Renew

Expiry Date

2025/05/12 02:28:34

Acquired DNS

192.168.126.2

Default gateway

192.168.126.2

Retrieve default gateway from server

Distance

5

Override internal DNS

Administrative Access

IPv4

☒ HTTPS

☐ FMG-Access

☐ FTM

☐ Speed Test

☒ HTTP

☒ SSH

☐ RADIUS Accounting

☒ PING

☐ SNMP

☐ Security Fabric Connection

Receive LLDP

Use VDOM Setting

Enable

Disable

Traffic Shaping

Outbound shaping profile

Miscellaneous

Comments

0/255

Status

Enabled

Disabled

OK

Cancel

The image above shows the first step in the firewall configuration which is to configure a wan interface, a alias was given with a role of wan interface, address was DHCP, for my home router which it got 192.168.126.134, administrator access was given https, ssh, ping.

Edit Interface

Name Lan (port2)

Alias

Type Physical Interface

VRF ID ⓘ

Role ⓘ

Address

Addressing mode Manual DHCP Auto-managed by IPAM

IP/Netmask

Create address object matching subnet ☒

Name port2 address

Destination

Secondary IP address ☐

Administrative Access

IPv4 ☐ HTTPS ☒ PING ☐ FMG-Access
☐ SSH ☐ SNMP ☐ FTM
☐ RADIUS Accounting ☐ Security Fabric Connection ⓘ ☐ Speed Test

Receive LLDP ⓘ Use VDOM Setting Enable Disable

Transmit LLDP ⓘ Use VDOM Setting Enable Disable

☒ DHCP Server

DHCP status + Enabled - Disabled

Address range

Netmask

Default gateway Same as Interface IP Specify

DNS server Same as System DNS Same as Interface IP Specify

Lease time ⓘ ☒ second(s)

Advanced

Network

Device detection ⓘ ☒

Security mode ☐

Traffic Shaping

Outbound shaping profile ☐

Miscellaneous

☒ DHCP Server

DHCP status Enabled Disabled

Address range

Netmask

Default gateway Same as Interface IP Specify

DNS server Same as System DNS Same as Interface IP Specify

Lease time 604800 second(s)

Network

Device detection ☒

Security mode ☐

DHCP was a requirement for the network so the lan link of port 2 acted as a dhcp server, status was enabled and address range was 192,168.10.2/24 since 192.168.10.1 was the gateway, lease time was set to default of 604800.

Edit Policy

Name

Internet

Incoming Interface

Lan (port2)

Outgoing Interface

Wan (port1)

Source

all

Destination

all

Schedule

always

Service

ALL

Action

☒ ACCEPT
 ☐ DENY

Firewall/Network Options

NAT ☒

IP Pool Configuration Use Outgoing Interface Address Use Dynamic IP Pool

Preserve Source Port ☐

Protocol Options default

Security Profiles

AntiVirus ☐

Web Filter ☐

DNS Filter ☐

Application Control ☐

IPS ☐

File Filter ☐

SSL Inspection no-inspection

Logging Options

Log Allowed Traffic Security Events All Sessions

Generate Logs when Session Starts ☐

Capture Packets ☐

Comments 0/1023

Enable this policy ☒

the image above shows a necessary firewall policy for internet access since FortiGate doesn't allow lan networks to access the internet, a policy was created with incoming interface as lan and outgoing interface as wan, since segmentation was not a requirement of the project, source and destination was set to all, with all services and Nat enabled.

New AntiVirus Profile

Name: Multi-AGT_AntiVirus

Comments: Write a comment... 0/255

AntiVirus scan: ☒ **Block** **Monitor**

Inspected Protocols

- HTTP ☒
- SMTP ☒
- POP3 ☒
- IMAP ☒
- FTP ☒
- CIFS ☐

APT Protection Options

- Treat Windows executables in email attachments as viruses ☒
- Send files to FortiSandbox for inspection ☐
- Include mobile malware protection ☒
- Quarantine ☐

Virus Outbreak Prevention

- Use FortiGuard outbreak prevention database ☐
- Use external malware block list ☐
- Use EMS threat feed ☐

OK **Cancel**

The image above shows the configuration of the antivirus policy, inspected protocols are http,smtp,pop3, imap, and ftp, while treating windows executables received in email addresses as virus as requested by the company, virus outbreak prevention was disabled since it uses a lot of Physical resources on the firewall.

New Web Filter Profile

Name

Multi-AGT_WebFiltering

Comments

Write a comment... 0/255

FortiGuard Category Based Filter

Warning: This device is not licensed for the FortiGuard web filtering service.

Traffic may be blocked if this option is enabled.

Allow

Monitor

Block

Warning

Authenticate

Name	Action
Local Categories 2	
custom1	Allow
custom2	Allow
Potentially Liabile 12	
Drug Abuse	Monitor
Hacking	Monitor
Illegal or Unethical	Monitor
Discrimination	Monitor
Explicit Violence	Monitor

0% 95

Allow users to override blocked categories

Static URL Filter

Block invalid URLs

URL Filter

Block malicious URLs discovered by FortiSandbox

Content Filter

Rating Options

Allow websites when a rating error occurs

Rate URLs by domain and IP Address

Proxy Options

HTTP POST Action

Allow

Block

Remove Cookies

✓ Allow 👁 Monitor 🚫 Block ⚠ Warning 👤 Authenticate	
Name	Action
Drug Abuse	👁 Monitor
Hacking	🚫 Block
Illegal or Unethical	👁 Monitor
Discrimination	👁 Monitor
Explicit Violence	👁 Monitor
Extremist Groups	🚫 Block
Proxy Avoidance	👁 Monitor
Plagiarism	👁 Monitor
Child Sexual Abuse	🚫 Block
8% 95	

 Allow	 Monitor	 Block	 Warning	 Authenticate
Name		Action		
Alternative Beliefs		 Warning		
<u>Abortion</u>		 Warning		
Other Adult Materials		 Warning		
Advocacy Organizations		 Warning		
Gambling		 Warning		
Nudity and Risque		 Warning		
Pornography		 Warning		
Dating		 Warning		
Weapons (Sales)		 Warning		
				21% 95

The image above shows the configuration of the web fileting policy for Multi Agt, FortiGuard category-based filter was enabled which is a database of websites provided for FortiGate users, which are divided by categories, with different actions, some changes were made based on company requirements, and invalid URLs were blocked.

Create New File Filter Rule

Name

Block EXE

Comments

Write a comment... 0/255

Protocols ⓘ

CIFS

FTP

HTTP

IMAP

POP3

SMTP

+

×

×

×

×

×

×

Traffic

Incoming

Outgoing

Both

Match Files

Password-protected only

File types

exe

+

×

Action

👁

Monitor

⊖

Block

OK

Cancel

New File Filter Profile

Name

Block_EXE

Comments

Write a comment... 0/255

Scan archive contents

☒

Rules

+ Create New

Edit

Delete

Search

Rule	Comments	Traffic	Protocols	Match Files	Action	File Types
Blo...		Both	HTTP IMAP POP3 SMTP +4	Any	Block	exe

1

The image above shows how to disable any computer under the firewall from downloading any exe files since they are not needed in the company and are risky, firewall filter rule was created using the name block exe, and in the match file, file type was set to exe, and action to block

Add Signatures

Type

Filter

Signature

Action

Default

Packet logging

Enable

Disable

Status

Enable

Disable

Default

Filter

os Windows

+

windows

Name	Severity	Target	OS	Action	CVE-ID
IPS Signature 4452/5295					
3Com.3CDaemon.FTP.Server.Buffer.Overflow	■■■■■	Server	Windows	Block	CVE-2005-0277
3Com.3CDaemon.FTP.Server.Information.Disc	■■■■■	Client	Windows	Pass	CVE-2005-0278
3Com.Intelligent.Management.Center.Informati	■■■■■	Server	Windows	Block	
3S.Pocknet.VMS.ActiveX.Control.Buffer.Overfl	■■■■■	Client	Windows	Block	CVE-2014-9263
3ivx.MPEG4.File.Processing.Buffer.Overflow	■■■■■	Client	Windows	Block	CVE-2007-6401
ABBS.Audio.Media.Player.LST.Buffer.Overflow	■■■■■	Server Client	Windows	Block	
ACal.Calendar.Cookie.Based.Authentication.By	■■■■■	Server	Windows Linux BSD Solaris MacOS	Block	CVE-2006-0182
AJDating.Viewprofile.PHP.SQL.Injection	■■■■■	Server	Windows Linux BSD Solaris MacOS	Block	CVE-2007-1297
ALLPlayer.M3U.Buffer.Overflow	■■■■■	Client	Windows	Block	
AOL.ICQ.ActiveX.Control.Remote.Code.Executi	■■■■■	Client	Windows	Block	CVE-2006-5650 CVE-2006-5650
AOL.Picture.Editor.YGPPicEdit.DLL.ActiveX.Bu	■■■■■	Client	Windows	Block	
AOL.Radio.ActiveX.Remote.Stack.Overflow	■■■■■	Client	Windows	Block	CVE-2007-5755 CVE-2007-6250
APDF.Wav.To.MP3.Buffer.Overflow	■■■■■	Client	Windows	Block	
ASUS.Net4Switch.Ipswcom.ActiveX.Control.Sta	■■■■■	Client	Windows	Block	CVE-2012-4924
AT&T.Connect.Participant.Application.Buffer.O	■■■■■	Server Client	Windows	Block	CVE-2013-6029
AVS.Media.Player.ActiveX.setsource.Method.Cc	■■■■■	Client	Windows	Block	
AWStats.Configdir.Command.Execution	■■■■■	Server	Windows	Block	CVE-2005-0116

0% 4,452/5,295

OK

Cancel

Dashboard

Network

Policy & Objects

Security Profiles

Antivirus

Web Filter

DNS Filter

Application Control

Intrusion Prevention

File Filter

SSL/SSH Inspection

Application Signatures

IPS Signatures

Web Rating Overrides

Web Profile Overrides

VPN

User & Authentication

System

Security Fabric

Log & Report

New IPS Sensor

Name

Multi-Agt-Intrusion Prevention

Comments

Runs on windows end points only 32/255

Block malicious URLs

IPS Signatures and Filters

Create New

Edit

Delete

Details	Exempt IPs	Action	Packet Logging
os Windows		Default	Disabled

1

Botnet C&C

Scan Outgoing Connections to Botnet Sites

Disable

Block

Monitor

OK

Cancel

The image above shows the configuration of the intrusion prevention system for multi agencies network, the signatures were chosen for windows only, since they are heavy on the FortiGate's physical recourses, what it does basically is check every packet leaving or entering the FortiGate, and runs it on a database of known attack signatures on the firewall, and acts accordingly to the rule some are blocked which have a high severity and others are allowed or monitored based on severity.

New Policy

Name

Lan port security measures

Incoming Interface

Lan (port2)

Outgoing Interface

Wan (port1)

Source

all

Destination

all

Schedule

always

Service

ALL

Action

ACCEPT

DENY

Firewall/Network Options

NAT

IP Pool Configuration

Use Outgoing Interface Address

Use Dynamic IP Pool

Preserve Source Port

Protocol Options

PROT default

Security Profiles

AntiVirus

Multi-AGT_AntiVirus

Web Filter

Multi-AGT_WebFiltering

DNS Filter

default

Application Control

default

IPS

Multi-Agt-Intrusion Preventior

File Filter

default

SSL Inspection

custom-deep-inspection

Decrypted Traffic Mirror

Logging Options

Log Allowed Traffic

Security Events

All Sessions

Generate Logs when Session Starts

Capture Packets

Comments

Write a comment...

0/1023

Enable this policy

Create New

Edit

Edit in CLI

Delete

Policy lookup

Search

Export

Interface Pair View

By Sequence

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes	Type
Internet	all	all	always	ALL	ACCEPT	Enabled	no-inspection	UTM	38.09 KB	Standard
Lan (port2) → Wan (port1)	all	all	always	ALL	ACCEPT	Enabled	Multi-AGT_AntiVirus Multi-AGT_WebFiltering default default Multi-Agt-Intrusion Prevention certificate-inspection block-see	UTM	0 B	Standard
Wan (port1) → Lan (port2)	all	all	always	ALL	ACCEPT	Enabled	Multi-AGT_AntiVirus Multi-AGT_WebFiltering default default custom-deep-inspection block-see	UTM	0 B	Standard

Impact

The image above shows the firewall policy created to run the security policies which were custom made for multi-agencies general trading network, source and destination were set to all, requirements not needed by the company were set to default but still running as a security measure, with ssl inspection set to custom-deep-inspection since it does not

require the installation of certificates on end devices, and logging was set to log only security events for resource conservation.

5. Test Plan

The network infrastructure of Multi Agencies General Trading was evaluated with a focus on overall system reliability, security, and failover processes. Testing was scheduled after midnight to avoid disruption to business operations.

To assess redundancy and failover effectiveness, the primary ISP connection was manually disconnected. The response time of the backup ISP was measured to ensure a smooth transition with minimal downtime. Connectivity during the switch-over was monitored using ping tests, confirming that critical services remained operational.

Firewall security and threat prevention mechanisms were also evaluated to ensure complete protection. Web filtering and DNS filtering were tested by attempting to access restricted categories such as social media, streaming sites, and malicious websites. DNS filtering was assessed by trying to access blocked domains. The Intrusion Prevention System (IPS) was tested by simulating network attacks like port scans and exploit attempts to verify its detection and automatic blocking capabilities. Application control was tested by attempting to download restricted file types such as EXE, ZIP, and torrent files, ensuring the firewall enforced security policies effectively.

Email security was another focus area, with tests designed to evaluate the firewall's phishing and spam filtering abilities. Phishing emails and malware-laden attachments were sent from a controlled environment to determine if the firewall could correctly identify and block such threats. Further investigation into the filtering system ensured that valid business emails were not incorrectly classified.

Bandwidth and latency tests were conducted to assess overall network performance. Stress testing with high traffic loads was performed to evaluate how the firewall and ISP handled network congestion. These tests ensured that security measures did not degrade performance and that the infrastructure remained stable under heavy load.

All tests confirmed that the redundancy, filtering, and network security mechanisms worked as intended, ensuring a reliable and secure infrastructure for the business.

5.1. Acceptance Test Process and Results

The network infrastructure underwent acceptance testing to verify the success of the implemented changes. The tests evaluated network security, redundancy, and overall system performance to ensure that the new configurations met operational needs.

To test the ISP failover, the primary connection was deliberately disconnected to allow the backup ISP to take over. The smooth transition and minimal downtime confirmed that the

failover mechanism functioned as expected, assuring the team that internet access would remain available in case of an outage.

Firewall security was tested through various methods. Web filtering and DNS filtering were evaluated by attempting to access restricted content, confirming that the firewall successfully blocked unwanted websites and domains. IPS testing involved simulating multiple attack scenarios to assess detection and mitigation responses. Application control was also tested by restricting the download of specific file types in compliance with company policies.

Mail security testing involved sending emails containing suspicious attachments and phishing URLs to assess the firewall's ability to filter such threats. The system effectively identified and blocked potential email-based breaches.

Network performance was further assessed through DHCP testing, where recently connected devices were monitored to ensure correct and automatic IP address assignment. The tests confirmed that the network was functioning efficiently, with no delays or IP address conflicts.

The testing results showed that the new network improvements and security policies enhanced protection, stability, and operational efficiency. The project's objectives were successfully achieved, leaving Multi Agencies General Trading with a more secure and resilient IT environment.

6. Discussion

6.1. System Functionality

In today's business landscape, maintaining a secure and efficient infrastructure is essential for safeguarding data and ensuring smooth operations. Multi Agencies General Trading required a robust system capable of enhancing security, managing traffic effectively, and maintaining constant network connectivity.

The primary objective of the project was to strengthen security and optimize network performance through the implementation of advanced firewall features. Web filtering and DNS filtering were deployed to regulate internet access and block unauthorized websites. Intrusion Prevention Systems (IPS) were set up to promptly detect and halt online threats. Additionally, application control was put in place to restrict certain file downloads and ensure compliance with company policies.

A failover mechanism for the ISP was introduced to boost network reliability, allowing for a seamless transition between the primary and backup internet connections during

disruptions. This measure ensured that crucial business applications and cloud services remained accessible. Furthermore, to prevent potential data breaches, the firewall's email security functionality was configured to screen for malicious attachments and phishing emails.

To validate the system's functionality, it underwent thorough testing. Failover testing confirmed the network's stability during ISP outages. IPS performance was assessed through simulated attack scenarios, while web filtering and DNS filtering were tested by attempting to access blocked domains. The DHCP system was also checked to ensure it accurately and automatically assigned IP addresses to devices.

In conclusion, the project successfully delivered a secure, efficient, and scalable network architecture for Multi Agencies General Trading. The enhancements ensured high availability, robust protection against cyber threats, and improved traffic management, supporting uninterrupted business operations.

6.2. Achieved Objectives

The project successfully achieved its core objectives of enhancing Multi Agencies General Trading's system resilience, network security, and connectivity. ISP failover procedures were implemented to ensure continuous internet connectivity, and advanced firewall configurations were established to improve security. Application control, DNS filtering, and web filtering were set up to manage network traffic, prevent unauthorized access, and enforce corporate policies.

Redundancy was incorporated into the network design to ensure reliability. By setting up a backup ISP connection, the system was able to automatically switch to the backup link in case of an outage, minimizing downtime and maintaining seamless operations. The firewall's real-time threat detection and blocking capabilities, alongside the IPS, provided an additional layer of protection. Furthermore, email security was enhanced to detect phishing attempts and filter out harmful communications.

Scalability was addressed by automating IP address allocation through a DHCP server, allowing the network to grow with future demands. The firewall's traffic management tools also optimized bandwidth usage, ensuring that critical business applications received prioritized access.

Objective	Status	Description
Ensure secure internal network communication	Achieved	Firewall rules and network segmentation were implemented to control and isolate traffic within the network.

Maintain reliable external connectivity	Achieved	ISP failover was configured to guarantee uninterrupted internet access.
Strengthen network security	Achieved	IPS, web filtering, and DNS filtering were put in place to prevent cyber threats.
Provide redundancy for high availability	Achieved	A secondary ISP link was added to enable automatic failover and ensure continuous connectivity.
Control network traffic and application access	Achieved	Application control was configured to prevent unauthorized downloads and access to applications.
Secure email communication	Achieved	Mail security features were configured to block phishing attempts and malware.
Automate IP address allocation	Achieved	A DHCP server was deployed to manage IP allocation efficiently and scale as needed.

TABLE 4 – ACHIEVED OBJECTIVES

6.3. Project Issues

The implementation of the new network infrastructure at Multi Agencies General Trading faced several challenges that required quick adjustments. One of the initial problems was sourcing the correct FortiGate firewall model locally. Despite clear requirements, finding a vendor with available stock took longer than anticipated, which delayed the setup timeline.

Once the firewall arrived, the installation process revealed another challenge. During the integration, the network briefly went offline, leading to an unexpected connectivity disruption. This required prompt troubleshooting to restore access and ensure that the existing configurations remained intact.

Moreover, the Yearly AMC took a while to be done so, the firewall purchase was delayed due to multiple approval steps, affecting the delivery schedule and postponing the configuration start date.

There were also minor issues during configuration, with some settings, such as DNS filtering and policy adjustments, not applying correctly the first time. These required manual adjustments and referencing the documentation to align the firewall rules with the company's needs.

These challenges emphasized the need for careful planning, supplier coordination, and problem-solving, particularly when working with live infrastructure in a production environment.

Issues and Limitations	Resolved Method
Delay in sourcing the correct firewall model locally	Contacted multiple suppliers and selected a nearby vendor with confirmed stock.
Network went down during firewall setup	Performed an immediate rollback and reconfigured step by step to prevent a full outage.
AMC Contract Took longer than expected	Coordinated with internal departments and adjusted the schedule accordingly.
Minor configuration mismatches during setup	Rechecked the documentation and manually updated policies.

TABLE 5 - ISSUES AND LIMITATIONS

6.4. Backup Plan

To ensure network stability and rapid recovery from unexpected issues, Multi Agencies General Trading has implemented a comprehensive backup strategy for its network infrastructure. All firewall configurations, including NAT rules, policies, and object groups, are securely stored in an encrypted offsite location, allowing the IT team to restore or modify settings without risking data loss.

Router settings and static IP plans are documented and saved in both digital and printed formats, offering flexibility in case of hardware failure or accidental resets. Backup files are regularly updated to reflect any changes made during maintenance or expansion.

Additionally, the entire network layout, including switch connections and firewall placement, is saved as a labeled diagram for easy reference. This visual guide is crucial for applying changes or troubleshooting, especially if new team members need to quickly familiarize themselves with the network structure.

Snapshots of the current network setup in the test environment are also retained as a fallback, enabling quick restoration in case testing errors affect live operations. While live backups from the FortiGate firewall are managed internally, an encrypted external copy is stored for disaster recovery purposes.

These strategies collectively ensure that Multi AGT's network can be swiftly restored in various scenarios, from equipment failure to cybersecurity incidents, minimizing operational disruptions.

6.5. Future Work

Looking ahead, Multi Agencies General Trading plans to implement a dedicated network monitoring system to enhance the visibility and performance of its internal infrastructure. This system will provide real-time insights into potential issues, such as traffic congestion, high latency on devices, hardware failures, or security incidents, enabling the IT team to act swiftly before these problems impact operations.

Tools like SolarWinds Network Performance Monitor are being considered, given their ability to track device health, monitor bandwidth usage, and identify unusual traffic patterns across the network.

On the Layer 2 side, the company is exploring the implementation of Rapid Spanning Tree Protocol (RSTP) to prevent switching loops and improve network stability, particularly as the network expands.

Security remains a primary concern, and the company plans to deploy a next-generation firewall with advanced capabilities. This will be complemented by the integration of automated threat detection systems that can adapt to modern cybersecurity challenges. These upgrades aim to strengthen access control and prevent breaches, safeguarding sensitive business operations in the region.

Additionally, Multi AGT intends to expand MPLS and QoS configurations to ensure critical business applications receive prioritized bandwidth, even during peak network usage. The adoption of Software-Defined Networking (SDN) is also being explored to enhance the management of future network expansions, offering increased flexibility and the ability to quickly roll out updates across a growing infrastructure.

These planned developments will fortify Multi AGT's network, ensuring it remains scalable, secure, and efficient enough to meet the evolving business needs and challenges of the future.

6.6. My Experience

Designing and implementing the network for Multi Agencies General Trading was a challenging but rewarding experience. Building the network from scratch required not only technical expertise but also patience, adaptability, and problem-solving skills at every stage, from the initial design to the final testing and troubleshooting.

One of the key learning experiences came from dealing with setbacks like temporary network downtime during the firewall setup and delays in sourcing the correct FortiGate model locally. These challenges tested my ability to remain calm and focused while seeking solutions. I also faced technical issues such as minor misconfigurations and the pressure of revisiting systems to ensure stability, but these experiences encouraged me to dig deeper and find more efficient ways to maintain the network's reliability.

Throughout this process, I refreshed my understanding of fundamental concepts such as Layer 2 switching and deepened my knowledge of advanced topics like IPsec, ACLs, VRF, and dynamic routing protocols. Configuring redundancy, setting up the firewall from scratch, and optimizing internal connectivity all contributed to strengthening my practical and theoretical networking skills. I also recognized the importance of thorough documentation and secure backups, particularly when changes need to be reversed or reviewed.

Beyond the technical aspects, this project helped me develop a structured approach to overcoming unexpected challenges. It boosted my confidence in building secure, scalable, and reliable networks and reaffirmed my passion for networking and IT as I move forward in my career.

6.7. Bahraini Perspective

The network upgrade project at Multi Agencies General Trading directly supports Bahrain's digital transformation and economic development goals outlined in Vision 2030. By improving the IT infrastructure of a wholesale company that serves a public market, this initiative strengthens both the private sector and the everyday lives of people in Bahrain.

In a country like Bahrain, where small and medium-sized enterprises are pivotal to the economy, ensuring reliable connectivity between warehouses, suppliers, markets, and customers is essential for seamless business operations. A stable, secure, and scalable network enables better inventory management, faster communication between departments, and more efficient coordination with retailers and vendors across the island.

With the improved IT infrastructure, Multi AGT's IT department now plays a crucial role in not just addressing technical issues but also safeguarding company data, protecting customer information, and supporting daily business activities. Thanks to enhanced firewall protection, secure remote access, and regular backups, the company is better equipped to defend against cyber threats and poised for future digital growth.

7. LESPI

7.1. Legal

Multi Agencies General Trading (MULTI AGT) ensures full compliance with Bahrain's telecommunications laws and data protection regulations. The company adheres to the guidelines set by the Telecommunications Regulatory Authority (TRA) and follows the Personal Data Protection Law (PDPL), which emphasizes user consent, data confidentiality, and secure handling of sensitive business and customer information. The IT team at MULTI AGT ensures that all systems, including the firewall, backup ISP, and Azure Active Directory integration, comply with these legal standards and are documented accordingly.

7.2. Ethical

Ethically, MULTI AGT prioritizes transparency and trust in its IT management. The company responsibly collects and stores data, taking proactive steps to prevent unauthorized access or misuse. The implementation of cybersecurity measures such as next-generation firewalls and automated threat detection systems reflects a strong ethical commitment to protecting internal users and business partners, both within Bahrain and internationally.

7.3. Social

As a wholesale company integral to Bahrain's supply chain, MULTI AGT plays a crucial role in ensuring the smooth flow of products, communication, and logistics. A secure and efficient IT infrastructure is essential to maintaining business operations, benefiting not just the company but also the customers and businesses it supports. The IT department is vital in sustaining service reliability and customer satisfaction, contributing to a stronger local economy and community.

7.4. Professional

The IT department at MULTI AGT adheres to high professional standards by implementing industry best practices and modern technologies to manage the company's infrastructure. From setting up backup ISP routers to configuring centralized access through Azure Active Directory, every step of the process reflects meticulous planning, documentation, and execution. Regular updates, clear communication, and rapid responses to issues are integral to maintaining a robust and professional IT environment that supports the company's ongoing growth and success.

8. Conclusion

The successful planning and implementation of Multi Agencies General Trading's upgraded network infrastructure represents a significant milestone in creating a secure, reliable, and scalable system to support the company's current operations and future expansion. This project addressed critical networking needs, including stable communication across departments and with external partners, enhanced cybersecurity through firewalls and backup systems, and improved operational efficiency through technologies such as Azure Active Directory and a secondary ISP connection.

Throughout the process, various challenges arose—from revisiting foundational concepts to troubleshooting advanced configurations. Overcoming these challenges not only enhanced technical skills but also underscored the importance of flexibility, thorough planning, and precise documentation, particularly in high-stakes environments like the wholesale industry, which relies on seamless connectivity with market networks.

This initiative contributes to Bahrain's broader digital transformation goals under Vision 2030, fostering innovation, business continuity, and digital security. The upgraded IT infrastructure at MULTI AGT is not only capable of meeting current operational demands but is also equipped for future growth, ensuring smooth services for clients, partners, and the community.

In conclusion, this thesis highlights the integration of technology with legal compliance, ethical responsibility, and professional standards. It also sets the foundation for future upgrades, such as the incorporation of advanced monitoring tools and scalability features, which will keep MULTI AGT at the forefront of reliable digital operations in Bahrain's evolving business landscape.

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10. Abstract

Questionnaire

1: What are the main security concerns or threats you want the firewall to address?

- 2: How many users and devices are currently connected to the network?
- 3: Do you require VPN access for remote users? If yes, how many users?
- 4: What kind of internet services or applications should be allowed or blocked?
- 5: Is there a need to create separate security zones (e.g., for cameras, servers, guests)?
- 6: Do you need bandwidth control or Quality of Service (QoS) for certain services?
- 7: Are there specific ports or protocols that need to be opened or restricted?
- 8: Do you want firewall logging and reporting for network traffic and threat detection?
- 9: Is there an existing firewall solution in place? If yes, what are its limitations?
- 10: Would you like integration with Active Directory or user-based policies?